

NADHAPRIYAN M S

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PROFESSIONAL SUMMARY

Innovative and detail-oriented engineering enthusiast with hands-on skills in IoT systems, electric vehicle design, and 3D modelling using industry-standard tools such as AutoCAD, CATIA, SolidWorks, and Blender. Proficient in C, C++, and Python, with ongoing specialization in advanced Python and full-stack web development. Adept at bridging hardware and software through practical engineering solutions and eager to contribute to dynamic, technology-driven teams.

EDUCATION

Class 10- 72% | March 2020

Velammal Matriculation School , Paruthipattu.

Class 12- 70.67% | May 2022

Velammal Matriculation School , Paruthipattu.

B.E. Electronics and Communication Engineering
2022 - 2026
Rajalakshmi Institute of Technology

CGPA 7.1 (Up to 6th Semester)

EXPERIENCE

CLUX - STARTUP
RAION ENERGY TECH

- **Pioneered the design and manufacturing of innovative E-cycles**, combining lightweight architecture with energy-efficient performance tailored for urban mobility.
- **Developed a Cargo-cum-Commute Electric Cycle**, engineered for dual utility—seamlessly balancing last-mile delivery needs with everyday personal transportation.

ENGINEERING INTERN
CAMWARE TECHNOLOGIES LLP | (Ongoing)

- Electronics Manufacturing Services (EMS)
- Worked in the domain of Electronic System Design And Embedded Design (EMSDM).
- Working in the project - horse masking LED light assembly

TOOLS

- **Design & Simulation:**

  3D CAD Tools: AutoCAD, CATIA, SolidWorks

  3D Modelling & Visualization: Blender

  Design Simulation & Analysis: ANSYS, Fusion 360.

- **Programming & Development:**

  Languages: C, C++, Python (Advanced), HTML, CSS.

  Full Stack Development (Ongoing): Python

  Development Tools: VS Code, Git, GitHub.

- **Automation & Industrial:**

  IoT Platforms: Arduino IDE, Raspberry Pi.

  Microcontroller Programming: Arduino, ESP32.

  Sensors & Actuators: IR, Ultrasonic, Temperature, Relay modules.

  Simulation Tools: Tinkercad Circuits.

  E-Vehicle Systems: BLDC motor control, Battery Management Systems (BMS), Regenerative Braking Systems.

TECHNICAL SKILLS

- Programming Languages: Python (Advanced), C, C++,Verilog, MATLAB, Full Stack Development in Python (Ongoing).
- Embedded Systems & IoT: Arduino, ESP8266, ESP32,Raspberry Pi, Sensors Integration, IoT System Design
- Electronics & Digital Systems: Digital Electronics, Image Processing, Electronics Circuit Design, Debugging, PCB Design
- Web Development: HTML, CSS, Python

PROJECTS

Worked on various projects under these themes throughout my engineering journey

E-Vehicle, Robotics & Hardware Systems

AI-Driven Driverless Tractor

- Developed a driverless agricultural tractor equipped with AI algorithms, proximity sensors, and GPS guidance to enable autonomous navigation in farmlands.
- Focused on real-time obstacle detection, precision farming, and automation of traditional farming practices.

Perseverance-Inspired Crawling Robot (Fire Extinguisher Bot)

Research & Embedded Systems

HUPSS – Hanging Underwater Probes for Ships

- Engineered underwater research probes mounted on ships to monitor marine conditions for academic and ecological studies.
- Integrated waterproofed sensor housings and data logging systems for real-time aquatic research.

NeuroBridge – EEG-Based Brain-Computer Interface (On Research)

- Built a non-invasive BCI system to translate EEG brain signals into control commands, enabling communication for individuals with motor impairments.
- Used EEG headset, Python, and machine learning models for signal classification and interface interaction.

IoT & Smart Systems

IoT Smart Doorbell with Facial Recognition

- Designed a smart doorbell system with real-time facial recognition using OpenCV, ESP32, and cloud storage.
- Enabled secure access control and visitor logging through a mobile app interface.

Software & Cybersecurity

Digital Assistant for Legal Awareness & KYC Framework

- Developed a mobile app to provide legal guidance and facilitate Know Your Customer (KYC) documentation through a user-friendly interface.
- Built using Python, Flask, and integrated APIs for real-time legal updates.

Vulnerability Assessment of Institutional Web Portals

- Conducted a cybersecurity audit using Nessus to evaluate and report vulnerabilities in academic web systems.
- Recommended remediation strategies based on industry-standard CVSS scoring and risk impact analysis.

LEADERSHIP ACTIVITIES

- Founder of **CLUX** (Startup) – Mentored and guided (10 members total) in R&D and Manufacturing domain.
- Led 3 teams in mini-projects, Smart India Hackathon and Bharath Cycle Design Challenge (AICTE) .

ACHIEVEMENTS

- **Winner – Bharath Cycle Design Challenge (BCDC)**

Secured 1st place in the national-level cycle design competition organized by **AICTE**, for an innovative electric cycle concept emphasizing sustainability and modular utility . In the concept of Cargo cum Commute Electric Cycle .

- **Finalist – Smart India Hackathon (SIH) 2023**

Developed and presented a **Digital Assistant for Legal Awareness and KYC Framework**, selected for national-level evaluation under the software category.

- **Patent Filed – Cargo-cum-Commute Electric Cycle**

Successfully filed both a **Design patent** and **Utility / Prototype** for a dual-purpose E-cycle aimed at efficient cargo transport and personal commuting.

- **Runner-Up – IDEX Defence Innovation Challenge**

Recognized for **HUPSS (Hanging Underwater Probes for Ships)** as a top-performing solution in underwater naval research category, acknowledged under the Ministry of Defence's innovation program.

PATENTS

- **Design Patent –** Cargo Cum Commute Electric Cycle.
- **Utility / Prototype Patent –** Cargo Cum Commute Electric Cycle.

CERTIFICATIONS

- Advanced Diploma in Python Programming (ADPP) – CSC Institute
- Full Stack Development In Python – GUVI
- IP Addressing and Subnetting – Network Geek
- Cyber Security And Ethical Hacking (Advanced) – GUVI (Ongoing)
- Hybrid Vehicle Design & Analysis – PumoTechnovation (Ongoing)

LANGUAGES

- English (Professional Working Proficiency)
- Tamil (Native Proficiency)
- Hindi (Private Course)